# Question Paper

## Physics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

Names of units of some physical quantities are given in List-I and their dimensions formulae are given in List-II. Match the correct pairs in the lists

| List-I | List-II | |
|---|---|---|
| *A.* Pa – s | (*i*) | $\left[L^2\ T^{-2}\ K^{-1}\right]$ |
| *B.* $NmK^{-1}$ | (*ii*) | $\left[MLT^{-3}\ K^{-1}\right]$ |
| *C.* $Jkg^{-1}\ K^{-1}$ | (*iii*) | $\left[ML^{-1}\ T^{-1}\right]$ |
| *D.* $Wm^{-1}\ K^{-1}$ | (*iv*) | $\left[ML^2\ T^{-2}\ K^{-1}\right]$ |

*A* *B* *C* *D*

○ (iv) (iii) (i) (ii)

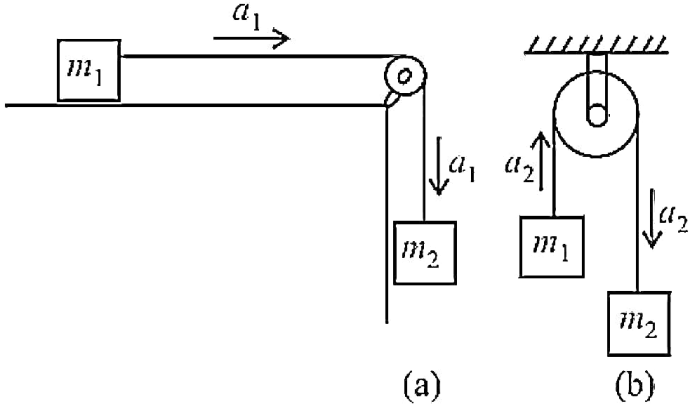
○ (iii) (ii) (iv) (i)

○ (iii) (i) (iv) (ii)

○ (iii) (iv) (i) (ii)

Question No. 2 **(Only One Option Correct Type)**

Two situations are shown in fig. (a) and (b).




In each case, $m_1 = 3$ kg and $m_2 = 4$ kg. If $a_1, a_2$ are the respective accelerations of the blocks in these situations, then the values of $a_1$ and $a_2$ are respectively [ $g = 10\ ms^{-2}$ ]

○ $\frac{20}{7}\ \mathrm{ms}^{-2}, \frac{10}{7}\ \mathrm{ms}^{-2}$

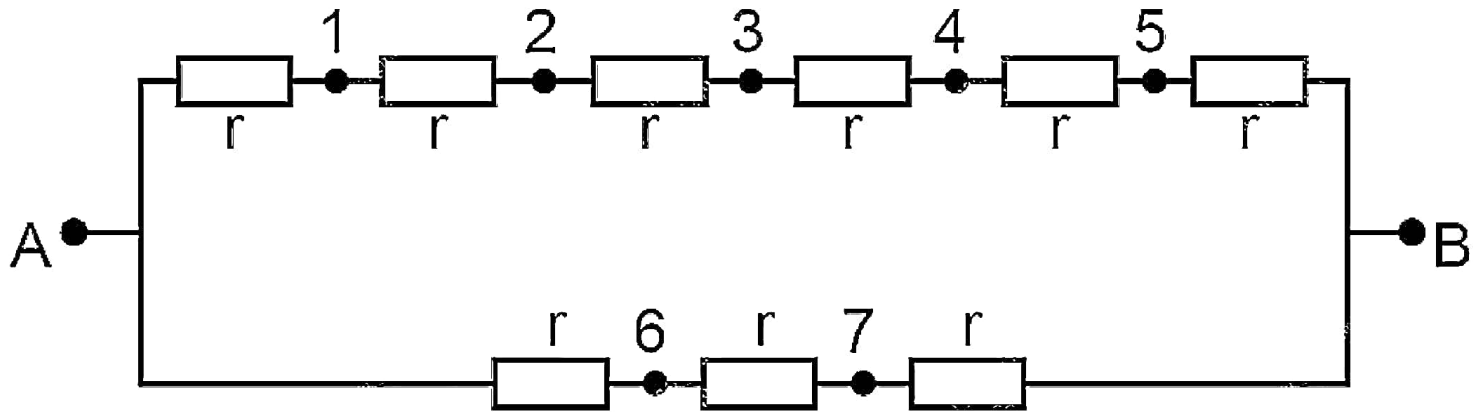
○ $\frac{10}{7}\ \mathrm{ms}^{-2}, \frac{25}{7}\ \mathrm{ms}^{-2}$

○ $\frac{40}{7}\ \mathrm{ms}^{-2}, \frac{10}{7}\ \mathrm{ms}^{-2}$

D) $\frac{30}{7}\mathrm{ms}^{-2}, \frac{5}{7}\mathrm{ms}^{-2}$

Question No. 3 **(Only One Option Correct Type)**

In the circuit shown in the figure all the resistance are identical and each has the value $r\Omega$. The equivalent resistance of the combination between the points $A$ and $B$ will remain unchanged even when the following pairs of points marked in the figure are connected through a resistance $\mathbb{R}$.




---

○ 2 and 6

○ 3 and 6

○ 4 and 7

○ Option (a) and (c)

Question No. 4 **(Only One Option Correct Type)**

The relation between $Y$, $\eta$ and $B$ is

○ $\frac{1}{\eta} = \frac{1}{3\eta} + \frac{1}{9B}$

○ $\frac{9}{Y} = \frac{1}{\eta} + \frac{3}{B}$

○ $\frac{1}{\eta} = \frac{1}{B} + \frac{1}{Y}$

○ $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{B}$

Question No. 5 **(Only One Option Correct Type)**

When a certain metallic surface was illuminated by light of 300 nm the stopping potential is found to 3 $V_0$. When a wavelength of 600 nm is incident on the surface the stopping potential is found to be $V_0$. Then the threshold wavelength for the metal would be: (in nm)

- 1200 nm
- 1600 nm
- 800 nm
- 1000 nm

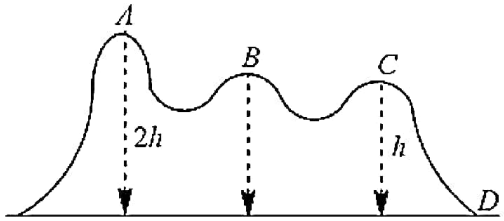
Question No. 6 **(Only One Option Correct Type)**

An electron in a hydrogen atom undergoes a transition from an orbit with a quantum number $n_i$ to another quantum number $n_f$ . $V_i$ and $V_f$ are the initial and final potential energies of the electron. If $\frac{V_i}{V_f} = 6.25$ then the smallest $n_f$ is

- 4
- 3
- 5
- 2

Question No. 7 **(Only One Option Correct Type)**

A small roller coaster starts at point A with a speed $u$ on a curved track as shown in the figure. The friction between the roller coaster and the track is negligible and it always remains in contact with the track. The speed of roller coaster at point $D$ on the track will be




- $(u^2 + gh)^{1/2}$
- $(u^2 + 2gh)^{1/2}$
- $(u^2 + 4gh)^{1/2}$
- u

Question No. 8 **(Only One Option Correct Type)**

The magnetic field in a plane electromagnetic wave is given as

$$\mathbf{B} = \left(3 \times 10^{-7}\ \mathrm{T}\right) \sin\left(3 \times 10^{4} x + 9 \times 10^{12} t\right) \hat{j}$$

The electric field of this wave is given as

- ○ $90 \sin\left(3 \times 10^{4} x + 9 \times 10^{12} t\right) \hat{i}\,\mathrm{Vm}^{-1}$
- ○ $90 \sin\left(3 \times 10^{4} x + 9 \times 10^{12} t\right) \hat{k}\,\mathrm{Vm}^{-1}$
- ○ $45 \sin\left(3 \times 10^{4} x + 9 \times 10^{12} t\right) \hat{i}\,\mathrm{Vm}^{-1}$
- ○ $45 \sin\left(3 \times 10^{4} x + 9 \times 10^{12} t\right) \hat{k}\,\mathrm{Vm}^{-1}$

Question No. 9 **(Only One Option Correct Type)**

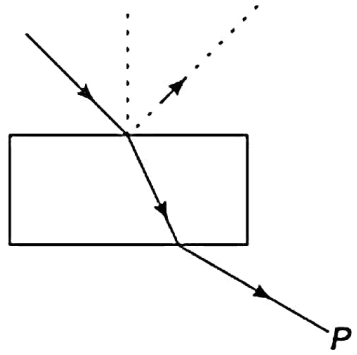
A circular coil connected to a battery of emf $E$ produced a certain magnetic induction field at its centre. The coil is unwound, stretched to double its length rewound into a coil of $\frac{1^{\text{rd}}}{3}$ of the original radius and connected to a battery of emf $E'$ to produce same field at the centre. Then $E'$ is

- ○ $\frac{2E}{9}$
- ○ $\frac{3E}{7}$
- ○ $\frac{9E}{4}$
- ○ $\frac{7E}{4}$

Question No. 10 **(Only One Option Correct Type)**

Consider a light beam incident from air to a glass slab at Brewster's angle as shown in the figure.

A polaroid is placed in the path of the emergent ray at a point $P$ and rotated about an axis passing through the centre and perpendicular to the plane of the polaroid.



A For a particular orientation there shall be darkness as observed through the polaroid

B The intensity of light as seen through the polaroid shall be independent of the rotation

C The intensity of light as seen through the polaroid shall go through a minimum but not zero for two orientations of the polaroid

D The intensity of light as seen through the polaroid shall go through a minimum for four orientations of the polaroid

Question No. 11 **(Only One Option Correct Type)**

The secondary coil of an ideal step down transformer is delivering 500 watt power at 12.5 A current. If the ratio of turns in the primary to the secondary is 5 : 1, then the current flowing in the primary coil will be :

○ 62.5 A

○ 2.5 A

○ 6 A

○ 0.4 A

Question No. 12 **(Only One Option Correct Type)**

A uniform solid sphere of radius $R$ produces a gravitational acceleration of $a_o$ on its surface. The distance of the point from the centre of the sphere where the gravitational acceleration becomes $\frac{a_o}{4}$ is,

○ $4R$

○ $\frac{3}{2}R$

○ $2R$

○ $3R$

Question No. 13 **(Only One Option Correct Type)**

Statement (I) : The temperature coefficient of resistance for most of metals in pure form is positive.

Statement (II) : A metal wire 2 mm in diameter carries a charge of $360\pi$C in two hours. If the metal contains $5 \times 10^{22}$ free electrons $/cm^3$, then drift velocity of the electrons in the wire is $6.25 \times 10^{-6}$ m/s.

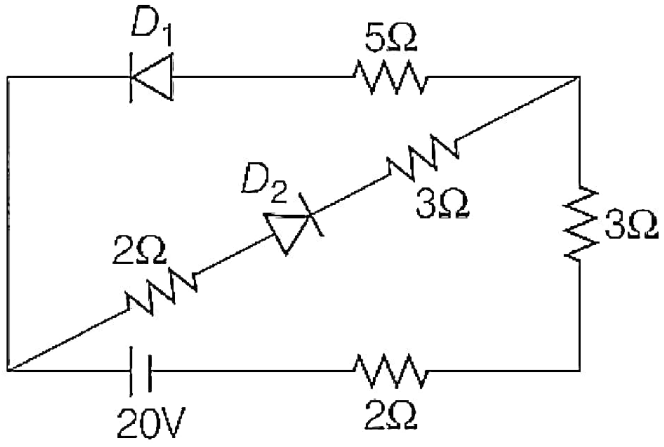
Statement (III) : Semiconductors like pure germanium does not obey Ohm's law for all range of electric field values. Which of the following is correct?

○ Statements I, II, III are true.

- Statements I, II are true, but statement III is false.
- Statements II, III are true, but statement I is false.
- Statements I, II, III are false.

Question No. 14 **(Only One Option Correct Type)**

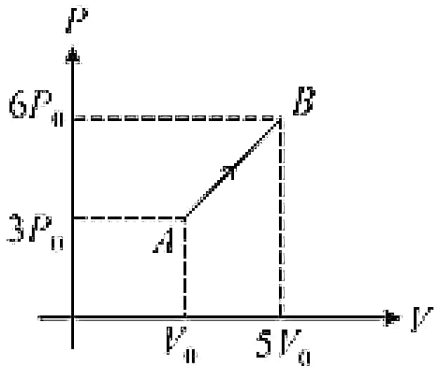
If the diodes are ideal in the circuit given below, then the current through the cell is




- 4 A
- 1.5 A
- 2 A
- 3 A

Question No. 15 **(Only One Option Correct Type)**

One mole of a monatomic ideal gas undergoes the process A $\rightarrow$ B in the given $p$ - $V$ diagram. Molar heat capacity in the process is




(a) $\frac{6R}{13}$

(b) $\frac{5R}{13}$

(c) $\frac{13R}{6}$

(d) $\frac{13R}{5}$

Question No. 16 **(Only One Option Correct Type)**

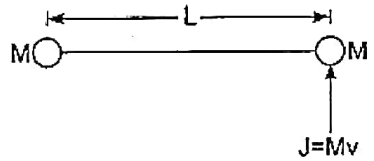
Assertion: No work is done in moving an electric dipole transnationally in uniform electric field.

Reason: Net force on electric dipole in uniform electric field is zero.

- ○ If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- ○ If Assertion is true but Reason is false.
- ○ If Assertion is false but Reason is true.
- ○ If both Assertion and Reason are false.

Question No. 17 **(Only One Option Correct Type)**

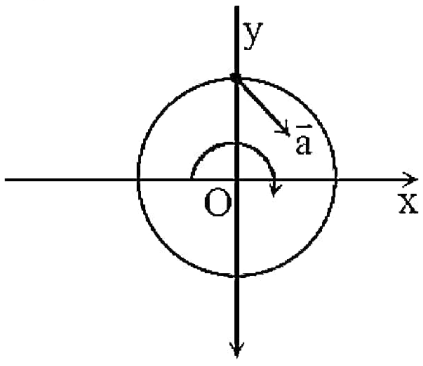
Consider a body, shown in diagram, consisting of two identical balls, each of mass $M$ connected by a light rigid rod. If an impulse $J = Mv$ is imparted to the body at one of its ends, what would be its angular velocity?




- ○ v/L
- ○ 2v/L
- ○ v/3L
- ○ v/4L

Question No. 18 **(Only One Option Correct Type)**

A body is moving is $x - y$ plane as shown in a circular path of radius 2 m . At a certain instant when the body is crossing the positive y -axis its acceleration is $\left(6\hat{i} - 8\hat{j}\right)\ \mathrm{m\,s^{-2}}$ . Then its angular acceleration and angular velocity at this instant will be-



- $-3\hat{k}\ \text{rad s}^{-2}$ and $-2\hat{k}\ \text{rad s}^{-1}$ respectively
- $+3\hat{k}\ \text{rad s}^{-2}$ and $+2\hat{k}\ \text{rad s}^{-1}$ respectively
- $-4\hat{k}\ \text{rad s}^{-2}$ and $-\sqrt{3}\,\hat{k}\ \text{rad s}^{-1}$ respectively
- $+4\hat{k}\ \text{rad s}^{-2}$ and $+\sqrt{3}\,\hat{k}\ \text{rad s}^{-1}$ respectively

Question No. 19 **(Only One Option Correct Type)**

When the temperature of an ideal gas is increased from $27^{\circ}\text{C}$ to $127^{\circ}\text{C}$, calculate the percentage increase in its $v_{rms}$.

- $37\%$
- $11\%$
- $33\%$
- $15.5\%$

Question No. 20 **(Only One Option Correct Type)**

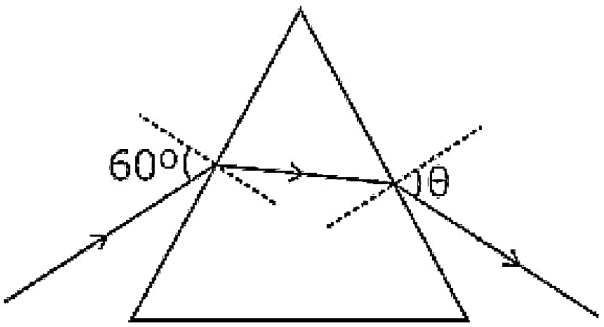
When an inductor $L$ and a resistor $R$ in series are connected across a $12\ \text{V}, 50\ \text{Hz}$ supply, a current of $0.5\ \text{A}$ flows in the circuit. The current differs in the phase from applied voltage by $\frac{\pi}{3}$ radian. Then the value of $R$ is

- $10\Omega$
- $3\Omega$
- $12\Omega$
- $15\Omega$

# Physics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

A monochromatic beam of light is incident at $60^{\circ}$ on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle $\theta(\mathrm{n})$ with the normal (see the figure). For $\mathrm{n} = \sqrt{3}$ the value of $\theta$ is $60^{\circ}$ and $\frac{d\theta}{dn} = \mathrm{m}$. The value of m is




Question No. 2 **(Numerical Type)**

A body travelling along a straight line path travels first half of the distance with a velocity $7\ \mathrm{ms}^{-1}$. During the travel time of the second half of the distance, first half time is travelled with a velocity $14\ \mathrm{ms}^{-1}$ and the second half time is travelled with a velocity $21\ \mathrm{ms}^{-1}$. Then the average velocity of the body during the journey is

Question No. 3 **(Numerical Type)**

Two identical discs are positioned on a vertical axis. The bottom disc is rotating with angular velocity $\omega_0$. The top disc is initially at rest. It is allowed to fall and sticks to the lower disc. Ratio of Kinetic Energy before and after collision is $M : 1$, find $M$.

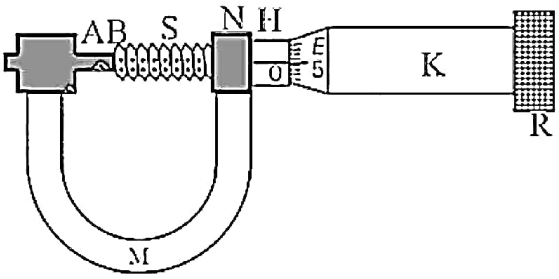
Question No. 4 **(Numerical Type)**

A closed pipe is suddenly opened and changed to an open pipe of same length. The fundamental frequency of the resulting open pipe is less than that of 3rd harmonic of the earlier closed pipe by 55 Hz. Then, the value of fundamental frequency of the closed pipe is

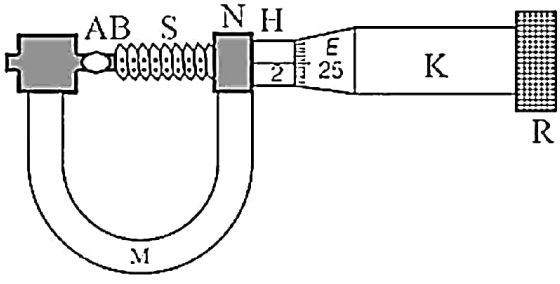
Question No. 5 **(Numerical Type)**

The circular scale of a screw gauge has 50 divisions and pitch of 0.5 mm.
If main scale reading is 2, then the diameter of sphere is ..... micrometer.

**(I)**




**(II)**



# Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

Which one of the following gives Prussian blue colour?

- $Fe_2[Fe(CN)_6]$
- $Na_4[Fe(CN)_6]$
- $Fe_3[Fe(CN)_6]_3$
- $Fe_4[Fe(CN)_6]_3$

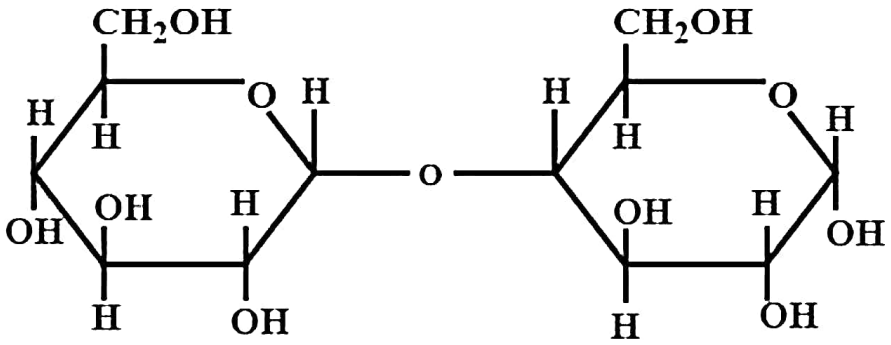
Question No. 2 **(Only One Option Correct Type)**

Alkyl cyanides undergo Stephen reduction to produce

- Aldehyde
- Secondary amine
- Primary amine
- Amide

Question No. 3 **(Only One Option Correct Type)**

Study the structure of maltose and mark the incorrect statement.




A. Maltose is composed of two α – D– glucose units
B. C – 1 of one glucose is linked to C – 4 of other unit
C. It is a non-reducing sugar
D. It is a disaccharide

Question No. 4 **(Only One Option Correct Type)**

How many of the following statements are correct?

(a) 'He' is the second most abundant element in the universe.

(b) the symbol for the element with atomic number 110 is Ds.

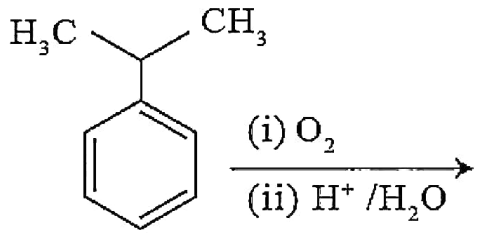
(c) Osmium has the highest density among all elements.

(d) Francium is the most electropositive element in the periodic table.

- ○ 3
- ○ 2
- ○ 4
- ○ 1

Question No. 5 **(Only One Option Correct Type)**

What by product is formed in the process of making phenol from cumene?




- ○ propan-2-ol
- ○ *n*-propanol
- ○ propanal
- ○ propan-2-one

Question No. 6 **(Only One Option Correct Type)**

Which of the following set of elements do not possess *f*-electrons?

- ○ La, U, Lr
- ○ La, Th, Lr
- ○ La, Ac, Th

○ Ce, Ac, Th

Question No. 7 **(Only One Option Correct Type)**

$[PdCl_2(PMe_3)_2]$ is a diamagnetic complex of $Pd(II)$. How many total isomers are possible for $[NiCl_2(PMe_3)_2]$ having oxidation state of Ni is $(II)$ ? (do not count given compound in isomers)

○ 3

○ 2

○ 1

○ Zero

Question No. 8 **(Only One Option Correct Type)**

Suppose the potential energy between electron and proton at a distance $r$ is given by $-\frac{Ke^2}{2r^3}$. Using Bohr's theory choose the correct statements.

○ Energy in the $n^{th}$ orbit is proportional to $n^2$

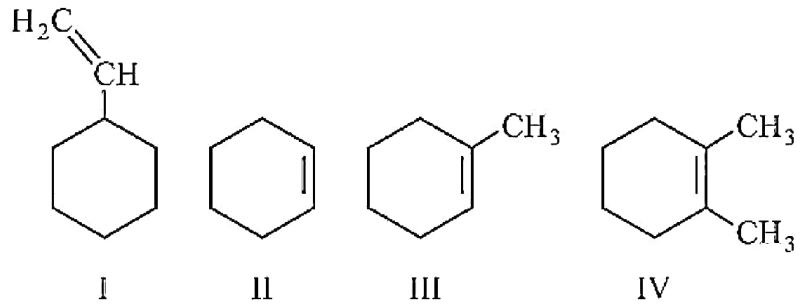
○ Energy is the $n^{th}$ orbit is proportional to $n^6$

○ Energy is proportional to $m^2$ ($m$: mass of electron)

○ Energy is proportional to $m^{-3}$ ($m$: mass of electron)

Question No. 9 **(Only One Option Correct Type)**

The correct order of the stability of the following compounds based on hyperconjugation is




I II III IV

○ IV > III > II > I

○ IV > II > I > III

○ IV > II > III > I

D) IV > I > III > II

Question No. 10 **(Only One Option Correct Type)**

Match the molecules given from (List I) with the shape obtained through VSEPR theory (List II).

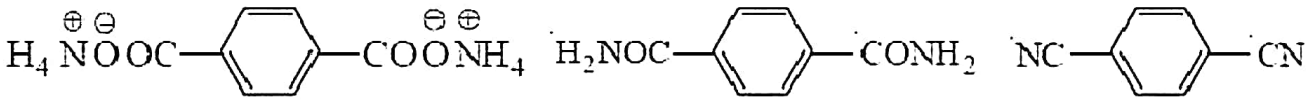
| | List (I) Molecules | | List (II) Shapes |
|---|---|---|---|
| A | $PCl_5$ | i | V-Shaped |
| B | $OF_2$ | ii | Triangular planar |
| C | $BCl_3$ | iii | Trigonal bipyramidal |
| D | $NH_3$ | iv | Trigonal pyramidal |
| | | v | Tetrahedral |

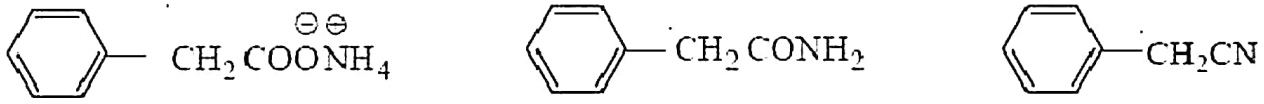
- A – i, B – v, C – iv, D – iii
- A – ii, B – iii, C – i, D – ii
- A – iv, B – iii, C – ii, D – v
- A – iii, B – i, C – ii, D – iv

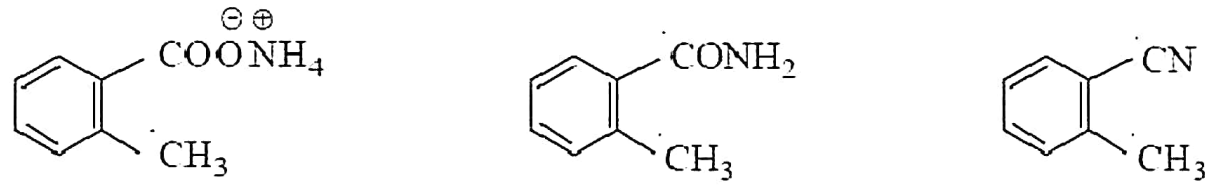
Question No. 11 **(Only One Option Correct Type)**

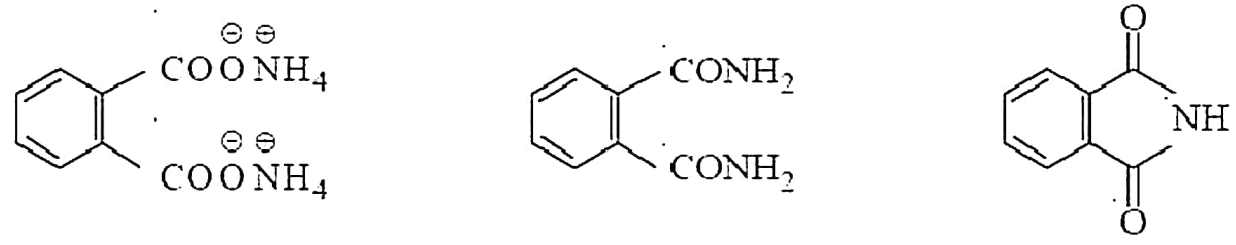
What are $A, B$ and $C$ in the following reactions?

$$\text{Phthalic acid} + NH_3 \longrightarrow A \xrightarrow{\Delta} B \xrightarrow[\text{temperature}]{\text{High}} C$$

1. 

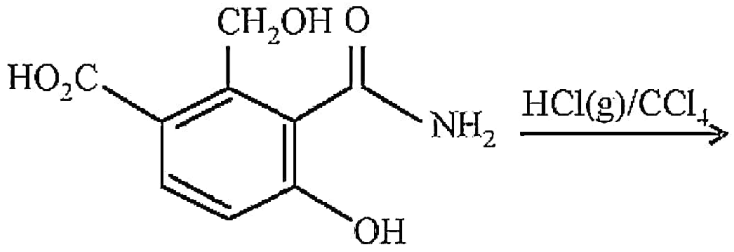
2. 

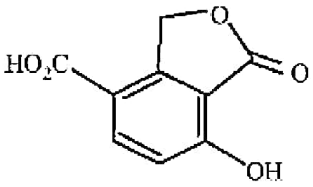
3. 

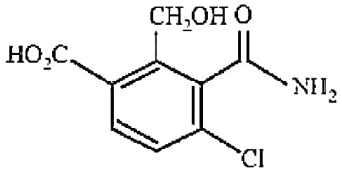
4. 

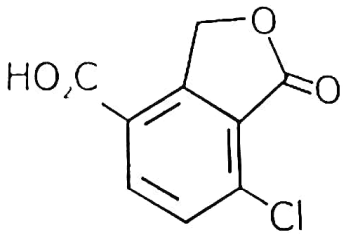
Question No. 12 **(Only One Option Correct Type)**

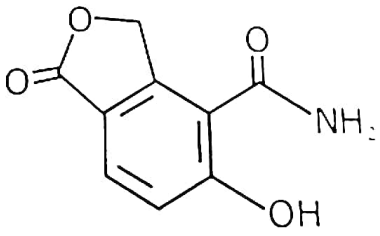
The major product expected from the following reaction is :




○ 


○ 


C) 


D) 


Question No. 13 **(Only One Option Correct Type)**

By diluting a weak electrolyte, specific conductivity ($K_c$) and equivalent conductivity ($\lambda_c$) change as-

○ Both increase

○ $K_c$ increases, $\lambda_c$ decreases

○ $K_c$ decreases, $\lambda_c$ increases

○ Both decrease

Question No. 14 **(Only One Option Correct Type)**

During the depression in freezing point experiment, an equilibrium is established between the molecules of

- liquid solvent and solid solvent
- liquid solute and solid solvent
- liquid solute and solid solute
- liquid solvent and solid solute

Question No. 15 **(Only One Option Correct Type)**

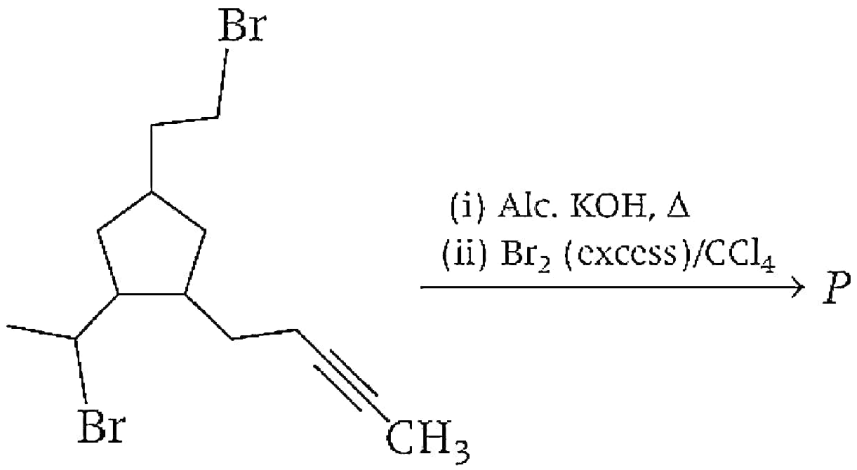
Match the following?

| property | metal |
|---|---|
| (a) Element with highest second ionization enthalpy ($\Delta_iH_2$) | (i) Co |
| (b) Element with highest third ionization enthalpy ($\Delta_iH_3$) | (ii ) Cr |
| (c) M in $[M(CO)_6]$ | (iii ) Cu |
| (d) Element with highest heat of atomization ($\Delta_aH$) | (iv ) Zn |
| | (v ) Ni |

- (a → iii ),(b → i ),( c → v ),( d → iv)
- (a → iii), (b → iv), (c → ii), (d → v)
- (a → iv), (b → i), (c → ii), (d → iii)
- (a → v), (b → iv), (c → i), (d → iii)

Question No. 16 **(Only One Option Correct Type)**

Maximum number of bromine atoms present in the final product $P$ upon complete bromination are



○ 4

○ 6

○ 10

○ 8

Question No. 17 **(Only One Option Correct Type)**

Which one of the following molecular formula cannot be represented by more than two types of functional group isomers?

○ $C_4H_8O$

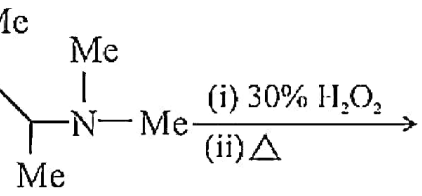
○ $C_4H_{10}O$

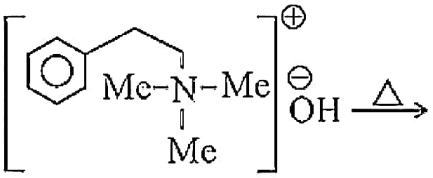
○ $C_3H_9N$

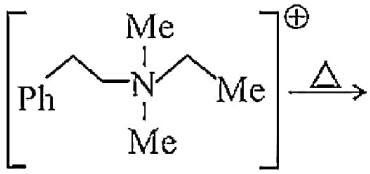
○ $C_4H_8O_2$

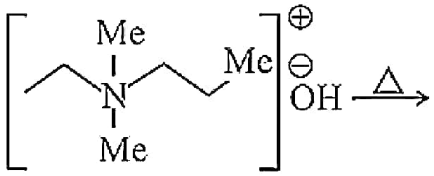
Question No. 18 **(Only One Option Correct Type)**

Which of the following would give hofmann alkene ?

(I) 


(II) 


(III) 


(IV) 


#no options available

Question No. 19 **(Only One Option Correct Type)**

$[X] + \text{dil. } H_2SO_4 \longrightarrow [Y]$ : Colourless, suffocating gas
$[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ Green colouration of solution
Then, $[X]$ and $[Y]$ are

- ○ $SO_3^{2-}, SO_2$
- ○ $Cl^-, HCl$
- ○ $S^{2-}, H_2S$
- ○ $CO_3^{2-}, CO_2$

Question No. 20 **(Only One Option Correct Type)**

Select the correct statement among the following.

- ○ Co (III) is stabilised in presence of weak field ligands, while Co (II) is stabilised in presence of strong field ligand.
- ○ Four coordinated complexes of Pd (II) and Pt (II) are paramagnetic and square planar.
- ○ $[Ni(CN)_4]^{4-}$ ion and $[Ni(CO)_4]$ are tetrahedral and square planar respectively.
- ○ $Ni^{2+}$ ion does not form inner orbital octahedral complexes in presence of weak ligand fields.

# Chemistry Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

For the first order reaction
$C_2H_4O_{(g)} \rightarrow CH_{4(g)} + CO_{(g)}$,
the initial pressure of $C_2H_4O_{(g)}$ is 80 torr and total pressure at the end of 20 minutes is 120 torr. The time needed for 75% decomposition of $C_2H_4$ would be:

Question No. 2 **(Numerical Type)**

The work done (in Cal) in adiabatic compression of 2 moles of an ideal monatomic gas by the constant external pressure of 2 atm starting from an initial pressure of 1 atm and an initial temperature of 300 K is:
$[R = 2cal/mol - K]$

Question No. 3 **(Numerical Type)**

An organic compound of molecular formula $C_3H_7N$ was analysed for nitrogen by Dumas method. Find the volume (in mL) of nitrogen evolved at NTP from 2 g of the substance.

Question No. 4 **(Numerical Type)**

A carbonyl compound of formula $C_9H_{10}O(A)$, which is a benzene derivative gives orange precipitate with 2,4 -D.N.P. and also gives yellow precipitate with $I_2$ in presence of aqueous NaOH. The total no. of isomers possible for 'A' are_____

Question No. 5 **(Numerical Type)**

Normality of a mixed solution of sulphuric acid and hydrochloric acid is 0.6 N. 20 mL of this solution gives 0.4305 g of AgCl on reacting with $AgNO_3$ solution. The strength of $H_2SO_4$ in g / L in the mixed solution is in g/L and round off to the nearest integer

# Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

The equation of the image of the pair of rays $y = |x|$ by the line $y = 1$, is

- $y = -|x| + 2$
- $y = |x| - 2$
- $y = |x| + 1$
- $y = |x| - 1$

Question No. 2 **(Only One Option Correct Type)**

For an increasing A.P. $a_1, a_2, \ldots . a_n$ if $a_1 + a_3 + a_5 = -12$ and $a_1 \cdot a_3 \cdot a_5 = 80$, then which of the following is incorrec

- $a_1 = -10$
- $a_2 = -1$
- $a_3 = -4$
- $a_5 = 2$

Question No. 3 **(Only One Option Correct Type)**

Let $z = x + iy$ be a complex number,
$A = \{z| \ |z| \leq 2\}$ and $B = \{z| \ (1-i)z + (1+i)\bar{z} \geq 4\}$
Then which one of the following options belongs to $A \cap B$ ?

- $\sqrt{3} + \frac{1}{2}\, i$
- $\frac{1}{2} + \frac{i}{2}$
- $\sqrt{2} + \frac{i}{2}$
- $2 + 2i$

Question No. 4 **(Only One Option Correct Type)**

Let the position vectors of points $A$, $B$ and $C$ in $\Delta ABC$ be $\hat{i}+\hat{j}+2\hat{k}$, $\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+\hat{j}+\hat{k}$, respectively. Let $l_1$, $l_2$ and $l_3$ be the lengths of the perpendiculars drawn from the orthocentre $O$ on the sides $AB$, $BC$ and $CA$ respectively. Then $(l_1+l_2+l_3)$ is equal to.....

- $\frac{4}{\sqrt{6}}$
- $\frac{\sqrt{6}}{2}$
- $\frac{3}{\sqrt{6}}$
- $\frac{\sqrt{6}}{5}$

Question No. 5 **(Only One Option Correct Type)**

The parabola $x^2=4ay$ makes an intercept of length $\sqrt{40}$ units on the line $y=1+2x$, then a value of $4a$ is

- 2
- -2
- -1
- 4

Question No. 6 **(Only One Option Correct Type)**

Let $f:R\to R$ such that $f(x+2y)=f(x)+f(2y)+4xy, x, y\in R$ and $f'(0)=0$
If $I_1=\int_0^1 f(x)dx, I_2=\int_{-1}^0 f(x)dx, I_3=\int_{\sqrt{1/2}}^2 f(x)dx$, then

---

- $I_1=I_2>I_3$
- $I_1>I_2>I_3$
- $I_1=I_2<I_3$
- $I_1<I_2<I_3$

Question No. 7 **(Only One Option Correct Type)**

Let $\alpha$, $\beta$ and $\gamma$ be the roots of $f(x)=0$, where $f(x)=x^3+x^2-5x-1$. Then the value of $[\alpha]+[\beta]+[\gamma]$ is equal to (where [·] denotes the greatest integer function)

- $-1$

○ 1

○ 4

○ $-3$

---

Question No. 8 **(Only One Option Correct Type)**

The area of the region bounded by the curve $C : y = \frac{x \div 1}{x^2 \div 1}$ and the line $L : y = 1$, is---

○ $1 - \frac{1}{2}\ln 2 + \frac{\pi}{4}$

○ $\ln 2 - \frac{\pi}{4} + 1$

○ $\frac{1}{2}\ln 2 \div \frac{\pi}{4} - 1$

○ $\ln 2 - \frac{\pi}{2} + 1$

Question No. 9 **(Only One Option Correct Type)**

Solution of differential equation

$$\left(1+e^{2y}\right)e^{\tan^{-1}x}dx - \left(1+x^2\right)\left(e^y + (e^y-1)^2\right)dy = 0$$

is-

(where C is constant of integration)----

○ $\ln(y) = \tan\left(y - e^{\tan^{-1}x} + C\right)$

○ $y = \ln\left(\tan\left(y - e^{\tan^{-1}x} + C\right)\right)$

○ $\ell\ln(y) = \tan\left(e^{\tan^{-1}x} - y + C\right)$

○ $y = \ln\left(\tan\left(e^{\tan^{-1}x} - y + C\right)\right)$

Question No. 10 **(Only One Option Correct Type)**

**Statement I:** The point $A(1,0,7)$ is the mirror image of the point $B(1,6,3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

**Statement II:** The line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ bisects the line segment joining $A(1,0,7)$ and $B(1,6,3)$.

- Statement I is true, statement II is true; statement II is not a correct explanation for statement I
- Statement I is true, statement II is false
- Statement I is false, statement II is true
- Statement I is true, statement II is true; statement II is a correct explanation for statement I

Question No. 11 **(Only One Option Correct Type)**

A man has to go from $(0,0)$ to $(8,8)$ by moving one step forward or upward on a $8 \times 8$ grid. If all paths are equally likely then the probability that he must pass through all the points of the form $(\alpha, \alpha^2) \& (\alpha, \alpha^3)$ is

- $\frac{1}{{}^{16}C_8}$
- $\frac{{}^2C_1 \cdot {}^4C_3 \cdot {}^4C_2 \cdot {}^6C_2}{{}^{16}C_8}$
- $\frac{{}^2C_1 \cdot {}^4C_3 \cdot {}^4C_2}{{}^{16}C_8}$
- $\frac{{}^2C_1 \cdot {}^4C_3}{{}^{16}C_8}$

Question No. 12 **(Only One Option Correct Type)**

If $[\cdot]$ denotes the greatest integer function, then the domain and range of the function $f(x) = \frac{\sin[x]\pi + \tan[x]\pi}{1+[x]^2 \div [x]^4}$ are respectively

- $\mathbb{R} - \{0\}, \mathbb{R} - \{0\}$
- $\mathbb{R}^+, \{0\}$
- $\mathbb{R}^+, \mathbb{R}$
- $\mathbb{R}, \{0\}$

Question No. 13 **(Only One Option Correct Type)**

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, a, b, \in N$. Then :

- there exist more than one but finite number of B's such that $AB = BA$
- there exists exactly one $B$ such that $AB = BA$

○ there exists infinitely many $B$ 's such that $AB = BA$

○ there cannot exist any $B$ such that $AB = BA$ C

Question No. 14 **(Only One Option Correct Type)**

The number of rational point(s) (a point $(a,\ b)$ is rational, if $a$ and $b$ both are rational numbers) on the circumference of a circle having centre $(\pi$,e) is

○ at most one

○ at least two

○ exactly two

○ infinite

Question No. 15 **(Only One Option Correct Type)**

$AB$ is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that $\triangle AOB$ (where ' $O$ ' is the origin) is an equilateral triangle, then the eccentricity e of the hyperbola satisfies

○ $e > \sqrt{3}$

○ $1 < e < \frac{2}{\sqrt{3}}$

○ $e = \frac{2}{\sqrt{3}}$

○ $e > \frac{2}{\sqrt{3}}$

Question No. 16 **(Only One Option Correct Type)**

The mean deviation from the median for the following data is

\begin{array}{ccccc} \hline \boldsymbol{x} & 10 & 11 & 12 & 13 \\ \hline \boldsymbol{f} & 6 & 12 & 18 & 12 \\ \hline \end{array}

○ S0.75S

○ S7.5S

○ S0.65S

○ S0.40S

#wtf?!!? MathonGo Suffering with latex errors now?

Question No. 16 **(Only One Option Correct Type)** **[FIXED]**

The mean deviation from the median for the following data is:

| $x$ | 10 | 11 | 12 | 13 |
|---|---|---|---|---|
| $f$ | 6 | 12 | 18 | 12 |

Options:

1. 0.75
2. 7.5
3. 0.65
4. 0.40

Question No. 17 **(Only One Option Correct Type)**

Let $A = \begin{bmatrix} 1 & a^3 & 0 \\ 0 & 1 & b^3 \\ c^3 & 0 & 1 \end{bmatrix}$ where $a, b, c \in R$. If the sum of all non-real roots of the equation $|A - xI| = 0$ is $k - mabc, \forall k, m \in Z$, then the value of $k + m$ is equal to

- 4
- 3
- 2
- 6

Question No. 18 **(Only One Option Correct Type)**

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} - 3\hat{k}$ be two vectors. If $A_1$ is the area of the quadrilateral having $\vec{a}, \vec{b}$ as its diagonals and $A_2$ is the area of the parallelogram having $\vec{a}$, $\vec{b}$ as its two adjacent sides, then $A_1 - A_2 =$

- 13
- 17
- 26
- 34

Question No. 19 **(Only One Option Correct Type)**

If $N$ is the number of positive integral solutions of the equation $x_1x_2x_3x_4 = 770$, then the value of $N$ is

- 250

○ 252

○ 254

○ 256

Question No. 20 **(Only One Option Correct Type)**

If $\text{Lim}_{n\to\infty} \frac{n\cdot 3^n}{n(x-2)^n + n\cdot 3^{n+1} - 3^n} = \frac{1}{3}$ then the range of $x$ is $(n \in N)$....

○ $[2, 5)$

○ $(1, 5)$

○ $(-1, 5)$

○ $(-\infty, \infty)$

## Mathematics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

Given two skew-lines with parametric equations $x = 1 + t,\ y = 1 + 6t,\ z = 2t$ and $x = 1 + 2s,\ y = 5 + 15s,\ z = -2 + 6s$. The shortest distance between these two given lines is equal to

Question No. 2 **(Numerical Type)**

Let $\mathrm{S} = \{1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7\}$. Then the number of possible functions $\mathrm{f} : \mathrm{S} \to \mathrm{S}$ such that $\mathrm{f}(\mathrm{m} \cdot \mathrm{n}) = \mathrm{f}(\mathrm{m}) \cdot \mathrm{f}(\mathrm{n})$ for every $\mathrm{m},\ \mathrm{n} \in \mathrm{S}$ and $\mathrm{m} \cdot \mathrm{n} \in \mathrm{S}$ is equal to _____.:

Question No. 3 **(Numerical Type)**

Given $f(x) = \begin{cases} x|x| & \text{for} \quad x \le -1 \\ [x+1] + [1-x] & \text{for} \quad -1 < x < 1 \\ -x|x| & \text{for} \quad x \ge 1 \end{cases}$, where [.] denotes the greatest integer function. If $I = \int_{-2}^{2} f(x)\,dx$, then $[3I] =$

Question No. 4 **(Numerical Type)**

If the number of terms free from radicals in the expansion of $\left(7^{\frac{1}{3}}+11^{\frac{1}{9}}\right)^{6561}$ is $k$, then the value of $k$ is equal to----

Question No. 5 **(Numerical Type)**

$\alpha, \beta$ be the roots of $x^2-3x+a=0$ and $\gamma, \delta$ be the roots of $x^2-12x+b=0$ and numbers $\alpha, \beta, \gamma, \delta$ (in order) form an increasing G.P. then the value of ab is

# (ANSWER KEY)

## Physics

| | | | | |
|---|---|---|---|---|
| Q1 [ ] | Q2 [ ] | Q3 [ ] | Q4 [ ] | Q5 [ ] |
| Q6 [ ] | Q7 [ ] | Q8 [ ] | Q9 [ ] | Q10 [ ] |
| Q11[ ] | Q12 [ ] | Q13 [ ] | Q14 [ ] | Q15 [ ] |
| Q16[ ] | Q17 [ ] | Q18 [ ] | Q19 [ ] | Q20 [ ] |

| | | | | |
|---|---|---|---|---|
| Q21 [ ] | Q22 [ ] | Q23 [ ] | Q24 [ ] | Q25 [ ] |

## Chemistry

| | | | | |
|---|---|---|---|---|
| Q1 [ ] | Q2 [ ] | Q3 [ ] | Q4 [ ] | Q5 [ ] |
| Q6 [ ] | Q7 [ ] | Q8 [ ] | Q9 [ ] | Q10 [ ] |
| Q11[ ] | Q12 [ ] | Q13 [ ] | Q14 [ ] | Q15 [ ] |
| Q16[ ] | Q17 [ ] | Q18 [ ] | Q19 [ ] | Q20 [ ] |

| | | | | |
|---|---|---|---|---|
| Q21 [ ] | Q22 [ ] | Q23 [ ] | Q24 [ ] | Q25 [ ] |

## Mathematics

| | | | | |
|---|---|---|---|---|
| Q1 [ ] | Q2 [ ] | Q3 [ ] | Q4 [ ] | Q5 [ ] |
| Q6 [ ] | Q7 [ ] | Q8 [ ] | Q9 [ ] | Q10 [ ] |
| Q11[ ] | Q12 [ ] | Q13 [ ] | Q14 [ ] | Q15 [ ] |
| Q16[ ] | Q17 [ ] | Q18 [ ] | Q19 [ ] | Q20 [ ] |

| | | | | |
|---|---|---|---|---|
| Q21 [ ] | Q22 [ ] | Q23 [ ] | Q24 [ ] | Q25 [ ] |